INCH-POUND
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12 June 2017
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# DETAIL SPECIFICATION

# CLOTH, DUCK, TEXTURED NYLON

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

## 1. SCOPE

1.1 Scope. This specification covers the requirements for textured nylon duck cloth, dyed or printed with a camouflage pattern.

1.2 Classification. The cloth covers the following types, classes and styles as specified:

Type I - 1000 denier
- Class 1 - 9.5 oz/sq yd, Untreated
- Class 2 - 9.5 oz/sq yd, Water repellent
- Class 3 - 12.0 oz/sq yd, Water repellent/back coated
- Class 4 - 12.0 oz/sq yd, Water repellent/flame retardant

Type II - 725 denier
- Class 1 - 7.5 oz/sq yd Untreated
- Class 2 - 7.5 oz/sq yd, Water repellent
- Class 3 - 10.0 oz/sq yd, Water repellent/back coated

Comments, suggestions, or questions on this document should be addressed to Attn: DLA Troop Support Standardization Team, 700 Robbins Avenue, Philadelphia, PA 19111-5096. Since contact information can change, you may want to verify the currency of the address information using Acquisition Streamlining and Standardization Information System (ASSIST) online database https://assist.dla.mil

AMSC NA FSC 8305

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Type III - 500 denier
- Class 1 - 7.0 oz/sq yd Untreated
- Class 2 - 7.0 oz/sq yd, Water repellent
- Class 3 - 8.0 oz/sq yd, Water repellent/back coated
- Class 4 - 9.5 oz/sq yd, Water repellent/flame retardant

Type IV - 330 denier
- Class 1 - 4.0 oz/sq yd Untreated
- Class 2 - 4.0 oz/sq yd, Water repellent
- Class 3 - 5.5 oz/sq yd, Water repellent/back coated

Style A - Solid shades
Style B - Woodland Camouflage printed
Style C - Desert Camouflage printed (3 Color)
Style D - Universal Camouflage printed (UCP)
Style E - Woodland Camouflage printed Marine Pattern (MARPAT)
Style F - Desert Camouflage printed, Marine Pattern (MARPAT)
Style G - Operation Enduring Freedom Camouflage Pattern (OEF-CP)
Style H - Operational Camouflage Pattern (OCP)

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-4 - Glossary of Fabric Imperfections

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1487 - Glossary of Cloth Coating Imperfections
(Copies are available online at http://quicksearch.dla.mil.)



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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY NATICK SOLDIER, RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

| | | |
|---|---|---|
| 2-1-1516 | - | Woodland Camouflage Pattern |
| 2-1-2240 | - | Desert Camouflage Pattern (3 color) |
| 2-1-2519 | - | Universal Camouflage Pattern |
| 2-1-2519-1 | - | ARPAT Camouflage Pattern, Desert Sand 500 |
| 2-1-2519-2 | - | ARPAT Camouflage Pattern, Urban Gray 501 |
| 2-1-2519-3 | - | ARPAT Camouflage Pattern, Foliage Green 502 |
| 2-1-2525 | - | Woodland MARPAT Pattern 4 color (Coyote 476) |
| 2-1-2526 | - | Woodland MARPAT Pattern 4 color (Green 474 with EGA symbol) |
| 2-1-2527 | - | Woodland MARPAT Pattern 4 color (Black 477) |
| 2-1-2528 | - | Woodland MARPAT Pattern 4 color (Khaki 475) |
| 2-1-2529 | - | Desert MARPAT Pattern 4 color (Light Tan 479) |
| 2-1-2530 | - | Desert MARPAT Pattern 4 color (Urban Tan 478) |
| 2-1-2531 | - | Desert MARPAT Pattern 4 color (Light Coyote 481 with EGA symbol) |
| 2-1-2532 | - | Desert MARPAT Pattern 4 color (Highland 480) |
| 2-1-2592 | - | Operational Camouflage Pattern |

(Copies of drawings are available from the U.S. Army Natick Soldier Research Development and Engineering Center, ATTN: RDNS-SEW-EWC, 10 General Greene Avenue, Natick, MA 01760-5019.)

CODE OF FEDERAL INSECTICIDE, FUNGICIDE AND RODENTICIDE ACT (FIFRA)

27 CFR Part 21 - Formula for Denatured Alcohol
40 CFR Parts (150 – 180) – Protection of Environment

(Copies are available on-line at http://www.gpoaccess.gov/cfr/index.html.)

ENVIRONMENTAL PROTECTION AGENCY

EPA chemicals-and-toxics-topics
Federal Insecticide, Fungicide and Rodenticide Act (FIFRA)
(Copies of these documents are available online at https://www.epa.gov/environmental-topics/chemicals-and-toxics-topics.)

FEDERAL TRADE COMMISION



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Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at http://www.ftc.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC Evaluation Procedure 1 – Gray Scale for Color Change
AATCC Evaluation Procedure 2 – Gray Scale for Staining
AATCC Evaluation Procedure 8 – AATCC 9-Step Chromatic Transference Scale
AATCC Evaluation Procedure 9 – Visual Assessment of Color Difference of Textiles
AATCC Test Method 8 - Colorfastness to Crocking: AATCC Crockmeter Method
AATCC Test Method 15 - Colorfastness to Perspiration
AATCC Test Method 16,2 - Colorfastness to Light: Carbon-Arc
AATCC Test Method 16.3 - Colorfastness to Light: Xenon-Arc
AATCC Test Method 20 - Fiber Analysis: Qualitative
AATCC Test Method 22 - Water Repellency: Spray Test
AATCC Test Method 61 - Colorfastness to Laundering, Accelerated
AATCC Test Method 70 - Water Repellency: Tumble Jar Dynamic Absorption Test
AATCC Test Method 81 - pH of the Water-Extract from Wet Processed Textiles
AATCC Test Method 96 - Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool
AATCC Test Method 118 - Oil Repellency: Hydrocarbon Resistance Test
AATCC Test Method 119 - Color Change Due to Flat Abrasion (Frosting): Screen Wire Method
AATCC Test Method 127 - Water Resistance: Hydrostatic Pressure Test
AATCC Test Method 135 - Dimensional Changes of Fabrics after Home Laundering
AATCC Test Method 169 - Weather Resistance of Textiles Xenon Lamp Exposure

(Copies of these documents are available on line at http://www.aatcc.org.)

ASTM INTERNATIONAL

ASTM D276 - Standard Test Methods for Identification of Fibers in Textiles
ASTM D737 - Test Method for Air Permeability of Textile Fabrics
ASTM D747 - Standard Test Method for Apparent Bending Modulus of Plastics by Means of a Cantilever Beam



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ASTM D751 - Standard Test Method for Coated Fabrics
ASTM D1424 - Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum Type (Elmendorf) Apparatus
ASTM D1776/D1776M - Standard Practice for Conditioning and Testing Textiles
ASTM D1907/D1907M - Standard Test Method for Linear Density of Yarn (Yarn Number) by the Skein Method
ASTM D2582 - Standard Test Method for Puncture-Propagation Tear Resistance of Plastic Film and Thin Sheeting
ASTM D3775 - Standard Test Method for Warp (End) and Filling (Pick) Count of Woven Fabrics
ASTM D3776/D3776M - Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3884 - Standard Guide for Abrasion Resistance of Textile Fabrics (Rotary Platform, Double-Head Method)
ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)
ASTM D6413/D6413M - Standard Test Method for Flame Resistance of Textiles (Vertical Test)

(Copies are available online at http://www.astm.org.)

OTHER PUBLICATIONS

Repeat Insult Patch Test-Modified Draize Procedure-
Principles and Methods of Toxicology, A Wallace Hayes (editor)

(Copies of these documents are available online at https://www.crcpress.com.)

Fabric Defect Replica Scales

(Fabric Defect Replica Scales are available online at http://www.sdlatlas.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Standard sample. The finished cloth shall match the standard sample for shade and appearance, and shall, be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).



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3.3 Recycled, recovered, or environmentally preferable or biobased materials. Recycled, recovered, or environmentally preferable or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Material.

3.4.1 Yarn. The yarn shall be textured continuous filament nylon. Testing shall be as specified in 4.4.5.

3.4.1.1 Yarn denier. The yarn denier shall be 1000 for Type I, 725 for Type II, 500 for Type III and 330 for Type IV. Testing shall be as specified in 4.4.5.

3.5 Color.

3.5.1 Style A, solid shades. The Style A finished cloth shall be dyed in one (1) of the following shades or as otherwise specified in the contract or purchase order: Camouflage Green 483, Foliage Green 504, Urban Gray 505. Coyote 498, Coyote Brown 3758, Tan 380, Tan 499 and Arctic White (see 6.2).

3.5.2 Style B, Woodland Camouflage print. The Style B finished cloth shall be dyed to a ground shade matching Light Green 354 and then shall be overprinted with the camouflage pattern. When the ground shade is dyed to match Light Green 354, the remaining three (3) colors shall be printed as appropriate, for the Dark Green 355, Brown 356, and Black 357 areas of the pattern. When the ground shade is not dyed to Light Green 354, all four (4) colors of the camouflage pattern shall be printed to match all four (4) colors of the pattern. The printing shall be accomplished by using a combination of acid dyes except for Black 357 only, carbon black alone or in combination with acid dyes shall be used.

3.5.3 Style C, Desert Camouflage print (UCP) (3 color). The Style C finished cloth shall be dyed to a ground shade matching Light Tan 492 and then overprinted with the camouflage pattern. When the ground shade is dyed to match Light Tan 492, the remaining two (2) colors shall be printed as appropriate, for the Light Brown 493 and Light Khaki 494 areas of the pattern. When the ground shade is not dyed to approximate Light Tan 492, all three (3) colors of the camouflage pattern shall be printed to match all three (3) colors of the pattern.

3.5.4 Style D, Universal Camouflage print (UCP). The Style D finished cloth shall be dyed to a ground shade matching Desert Sand 500 and then overprinted with the camouflage pattern. When the ground shade is dyed to match Desert Sand 500, the remaining two (2) colors shall be printed as appropriate, for the Urban Gray 501 and Foliage Green 502 areas of the pattern. When the ground shade is not dyed to Desert Sand 500, all three (3) colors of the camouflage pattern shall be printed to match all three (3) colors of the pattern.



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3.5.5 Style E, Woodland Camouflage print (MARPAT). The Style E finished cloth shall be dyed to a ground shade matching Khaki 475 and then overprinted with the camouflage pattern. When the ground shade is dyed to match Khaki 475, the remaining colors shall be printed as appropriate for the Green 474, Coyote 476 and Black 477 areas of the pattern. When the ground shade is not dyed to Khaki 475, all four (4) colors of the camouflage pattern shall be printed to match all four (4) colors of the pattern.

3.5.6 Style F, Desert Camouflage print (MARPAT). The Style F finished cloth shall be dyed to a ground shade matching Light Tan 479 and then overprinted with the camouflage pattern. When the ground shade is dyed to match Light Tan 479, the remaining colors shall be printed as appropriate for the Urban Tan 478, Highland 480 and Light Coyote 481 areas of the pattern. When the ground shade is not dyed to Light Tan 479, all four (4) colors of the camouflage pattern shall be printed to match all four (4) colors of the pattern.

3.5.7 Style G, Operation Enduring Freedom Camouflage Pattern (OEF-CP). The Style G finished cloth shall be dyed to a ground shade matching Cream 524 and then overprinted with the camouflage pattern. When the ground shade is dyed to match Cream 524, the remaining six (6) colors shall be printed as appropriate, for the Tan 525, Pale Green 526, Olive 527, Dark Green 528, Brown 529 and Dark Brown 530 areas of the pattern. When the ground shade is not dyed to Cream 524 all seven (7) colors of the camouflage pattern shall be printed to match all seven (7) colors of the pattern (see 6.4).

3.5.8 Style H, Operational Camouflage Pattern (OCP). The Style H finished cloth shall be dyed to a ground shade matching Dark Cream 559 and then shall be overprinted with the camouflage pattern. When the ground shade is dyed to match Dark Cream 559, the six (6) remaining colors shall be printed as appropriate, for the Tan 525, Light Sage 560, Olive 527, Dark Green 528, Brown 529 and Bark Brown 561 areas of the pattern. When the ground shade is not dyed to Dark Cream 559 all seven (7) colors of the camouflage pattern shall be printed to match all seven (7) colors of the pattern.

3.5.9 Visual shade matching. The color and appearance of the dyed or camouflage printed finished cloth shall match the standard sample when tested as specified in 4.4.5 (see 6.4).

3.5.10 Colorfastness. The finished cloth shall conform to the colorfastness requirements specified in Table I when tested as specified in 4.4.5.

TABLE I. Colorfastness requirements (all styles).



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| Style | Color Evaluation | Laundering Color Change & Staining (3 cycles) (min.) | Light (after 40 AFU or 170 kJ/ ($m^2$nm)@ 420 nm) (min.) 1/ | Perspiration (acid & alkaline) Color Change and Staining) (min.) | Crocking Dry/Wet (min.) | Accelerated weathering (80 AFU or 340 kJ/ ($m^2$nm)@ 420 nm) (min.) 1/, 2/ | Frosting (Carbon Black) (min.) |
|---|---|---|---|---|---|---|---|
| "A" Solid Shade | All colors | 3-4 | 3-4 | 3-4 | 3.5 | 3-4 | N/A |
| "B" Woodland Camouflage | All colors except Black 357 | 3-4 | 3-4 | 3-4 | 3.5 | N/A | N/A |
| | Black 357 | 3 | 2-3 | 3-4 | 1.0 | N/A | 3-4 |
| "C" Desert Camouflage 3 Color | All colors | 3-4 | 3-4 | 3-4 | 3.5 | N/A | N/A |
| "D" Universal Camouflage | All colors | 3-4 | 3-4 | 3-4 | 3.5 | N/A | N/A |
| "E" Woodland MARPAT | All colors except Black 477 | 3-4 | 3-4 | 3-4 | 3.5 | N/A | N/A |
| | Black 477 | 3-4 | 3-4 | 3-4 | 1.5 | N/A | 3-4 |

TABLE I. Colorfastness requirements (all styles). - Continued

| Style | Color Evaluation | Laundering Color Change & Staining (3 cycles) (min.) | Light (after 40 AFU or 170 kJ/ ($m^2$nm)@ 420 nm) (min.) 1/ | Perspiration (acid & alkaline) Color Change and Staining) (min.) | Crocking Dry/Wet (min.) | Accelerated weathering (80 AFU or 340 kJ/ ($m^2$nm)@ 420 nm) (min.) 1/, 2/ | Frosting (Carbon Black) (min.) |
|---|---|---|---|---|---|---|---|
| "F" Desert MARPAT | All colors | 3-4 | 3-4 | 3-4 | 3.5 | N/A | N/A |



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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| "G" OEF-CP Camouflage | Dk. Green 528, Brown 529, Dk. Brown 530 | 3-4 | 3-4 | 3-4 | 3.5 | N/A | N/A |
| | Cream 524, Tan 525 Pale Green 526, Olive 527 | 3-4 | 3 | 3-4 | 3.5 | N/A | N/A |
| "H" OCP Camouflage | Dk. Green 528, Brown 529, Bark Brown 561 | 3-4 | 3-4 | 3-4 | 3.5 | N/A | N/A |
| | Dark Cream 599, Tan 525, Light Sage 560, Olive 527 2/ | 3-4 | 3 | 3-4 | 3.5 | N/A | N/A |

1/ AFU: AATCC Fading Units
2/ Style A only

3.6 Pattern execution. The pattern on the printed finished cloth(s) shall match the standard sample with respect to design, colors, and registration of the respective areas. The various areas of the pattern shall be properly registered in relation to each other and shall present definite sharp demarcations with a minimum of feathering or spew. Each pattern area shall show solid coverage; skitteriness exceeding that shown by the standard sample in any of the printed areas will not be acceptable. When a standard sample is not available for pattern execution, a pattern drawing will be provided (see 6.4), and the pattern on the finished cloth shall match that of the drawing (see 2.2.2. and 6.2). The pattern repeat for each style shall be as follows:

Style B Woodland Camouflage - 27.25 (+1.25, -2.50) inches in the warp direction.
Style C Desert Camouflage (3 color) - 16.75 (+1.25, -2.25) inches in the warp direction.
Style D Universal Camouflage - 36.00 (+1.25, -2.50) inches in the warp direction.
Style E Woodland MARPAT - 27.25 (+1.25, -2.50) inches in the warp direction.
Style F Desert MARPAT - 27.25 (+1.25, -2.50) inches in the warp direction.
Style G Operation Enduring Freedom-Camouflage Pattern (OEF-CP) - 25.255 (+1.25,-2.50) inches in the warp direction.
Style H Operational Camouflage Pattern (OCP) - 25.255 (+1.25, -2.50) inches in the warp direction.

3.6.1 Pattern width (filling direction for Style H. The maximum camouflage printed area for Style H, shall be 68-inches in the filling direction.



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3.7 Spectral reflectance. The spectral reflectance values for each Style shall conform to the requirements specified in their respective applicable Tables II through IX when tested as specified in 4.4.5.

TABLE II. Spectral reflectance (percent), Style A.

| Solid Shades | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Wavelength (nanometers) | Camouflage Green 483 | | Foliage Green 504 | | Urban Gray 505 | | Coyote 498 | | Coyote Brown 3758 | | Tan 380 | | Tan 499 | |
| | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max |
| 600 | 3 | 10 | 8 | 26 | 12 | 26 | 8 | 20 | 8 | 20 | N/A | N/A | 8 | 26 |
| 620 | 3 | 10 | 8 | 26 | 14 | 26 | 8 | 20 | 8 | 20 | N/A | N/A | 8 | 26 |
| 640 | 3 | 10 | 8 | 28 | 14 | 28 | 8 | 22 | 8 | 22 | N/A | N/A | 8 | 30 |
| 660 | 3 | 11 | 10 | 30 | 14 | 30 | 8 | 24 | 8 | 26 | N/A | N/A | 8 | 34 |
| 680 | 3 | 13 | 10 | 34 | 18 | 34 | 12 | 24 | 10 | 27 | N/A | N/A | 12 | 38 |
| 700 | 4 | 28 | 12 | 38 | 24 | 38 | 12 | 34 | 12 | 53 | 25 | 53 | 12 | 40 |
| 720 | 5 | 40 | 16 | 42 | 26 | 42 | 16 | 42 | 16 | 54 | 25 | 54 | 16 | 46 |
| 740 | 7 | 52 | 16 | 46 | 30 | 46 | 22 | 46 | 20 | 55 | 25 | 55 | 22 | 50 |
| 760 | 11 | 60 | 18 | 48 | 32 | 48 | 30 | 50 | 21 | 56 | 26 | 56 | 30 | 50 |
| 780 | 17 | 64 | 18 | 48 | 34 | 48 | 34 | 54 | 21 | 57 | 27 | 57 | 34 | 54 |
| 800 | 24 | 67 | 20 | 50 | 34 | 50 | 36 | 56 | 22 | 58 | 28 | 58 | 36 | 56 |
| 820 | 32 | 70 | 22 | 54 | 36 | 54 | 38 | 58 | 23 | 59 | 30 | 59 | 38 | 58 |
| 840 | 37 | 71 | 24 | 54 | 38 | 54 | 38 | 58 | 24 | 62 | 33 | 62 | 38 | 58 |
| 860 | 40 | 73 | 26 | 56 | 40 | 56 | 40 | 60 | 25 | 65 | 36 | 65 | 40 | 60 |

TABLE III. Spectral reflectance (percent), Style B.

| Woodland Camouflage Pattern | | | | | | |
|---|---|---|---|---|---|---|
| Wavelengths (nanometers) | Light Green 354 | | Dark Green 355 & Brown 356 | | Black 357 | |
| | Min | Max | Min | Max | Min | Max |
| 600 | 8 | 20 | 3 | 9 | N/A | N/A |
| 620 | 8 | 20 | 3 | 9 | N/A | N/A |
| 640 | 8 | 20 | 3 | 9 | N/A | N/A |
| 660 | 8 | 20 | 3 | 12 | N/A | N/A |
| 680 | 10 | 30 | 3 | 16 | N/A | N/A |
| 700 | 18 | 50 | 5 | 32 | N/A | 20 |



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| Woodland Camouflage Pattern | | | | | | |
|---|---|---|---|---|---|---|
| Wavelengths (nanometers) | Light Green 354 | | Dark Green 355 & Brown 356 | | Black 357 | |
| 720 | 22 | 54 | 7 | 44 | N/A | 30 |
| 740 | 30 | 56 | 12 | 52 | N/A | 33 |
| 760 | 35 | 58 | 18 | 56 | N/A | 33 |
| 780 | 40 | 62 | 26 | 56 | N/A | 34 |
| 800 | 55 | 80 | 34 | 56 | N/A | 34 |
| 820 | 55 | 80 | 42 | 60 | N/A | 35 |
| 840 | 55 | 84 | 44 | 60 | N/A | 35 |
| 860 | 60 | 84 | 44 | 60 | N/A | 35 |

TABLE IV. Spectral reflectance (percent), Style C.

| Desert Camouflage Pattern (3-color) | | | | | | |
|---|---|---|---|---|---|---|
| Wavelength, (nanometers) | Light Tan 492 | | Light Brown 493 | | Light Khaki 494 | |
| | Min | Max | Min | Max | Min | Max |
| 700 | 38 | 53 | 19 | 36 | 25 | 48 |
| 720 | 38 | 58 | 20 | 36 | 25 | 52 |
| 740 | 39 | 62 | 20 | 36 | 25 | 54 |
| 760 | 40 | 66 | 21 | 36 | 26 | 56 |
| 780 | 41 | 72 | 21 | 38 | 27 | 57 |
| 800 | 43 | 76 | 22 | 43 | 28 | 58 |
| 820 | 45 | 76 | 23 | 45 | 30 | 58 |
| 840 | 48 | 78 | 24 | 46 | 33 | 58 |
| 860 | 50 | 78 | 25 | 46 | 36 | 59 |

TABLE V. Spectral reflectance (percent), Style D.

| Universal Camouflage Pattern | | | | | | |
|---|---|---|---|---|---|---|
| Wavelength, (nanometers) | Desert Sand 500 | | Urban Gray 501 | | Foliage Green 502 | |
| | Min | Max | Min | Max | Min | Max |
| 600 | 28 | 40 | 12 | 26 | 8 | 18 |
| 620 | 30 | 42 | 14 | 26 | 8 | 18 |
| 640 | 34 | 48 | 14 | 28 | 8 | 20 |
| 660 | 38 | 56 | 14 | 30 | 10 | 26 |
| 680 | 44 | 60 | 18 | 34 | 10 | 26 |
| 700 | 46 | 66 | 24 | 38 | 12 | 28 |



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| Universal Camouflage Pattern | | | | | | |
|---|---|---|---|---|---|---|
| Wavelength, | Desert Sand 500 | | Urban Gray 501 | | Foliage Green 502 | |
| 720 | 48 | 68 | 26 | 42 | 16 | 30 |
| 740 | 48 | 72 | 30 | 46 | 16 | 30 |
| 760 | 50 | 74 | 32 | 48 | 18 | 32 |
| 780 | 54 | 76 | 34 | 48 | 18 | 34 |
| 800 | 54 | 76 | 34 | 50 | 20 | 36 |
| 820 | 54 | 76 | 36 | 54 | 22 | 38 |
| 840 | 55 | 78 | 38 | 54 | 24 | 40 |
| 860 | 56 | 78 | 40 | 56 | 26 | 42 |

TABLE VI. Spectral reflectance (percent), Style E.

| Marine Pattern (MARPAT) Woodland Camouflage Pattern | | | | | | |
|---|---|---|---|---|---|---|
| Wavelengths (nanometers) | Khaki 475 & Coyote 476 | | Green 474 | | Black 477 | |
| | Min | Max | Min | Max | Min | Max |
| 600 | 8 | 20 | 3 | 9 | N/A | N/A |
| 620 | 8 | 20 | 3 | 9 | N/A | N/A |
| 640 | 8 | 20 | 3 | 9 | N/A | N/A |
| 660 | 8 | 20 | 3 | 12 | N/A | N/A |
| 680 | 10 | 30 | 3 | 16 | N/A | N/A |
| 700 | 18 | 50 | 5 | 32 | N/A | 20 |
| 720 | 22 | 54 | 7 | 44 | N/A | 30 |
| 740 | 30 | 56 | 12 | 52 | N/A | 33 |
| 760 | 35 | 58 | 18 | 56 | N/A | 33 |
| 780 | 40 | 62 | 26 | 56 | N/A | 34 |
| 800 | 55 | 80 | 34 | 56 | N/A | 34 |
| 820 | 55 | 80 | 42 | 60 | N/A | 35 |
| 840 | 55 | 84 | 44 | 60 | N/A | 35 |
| 860 | 60 | 84 | 44 | 60 | N/A | 35 |

TABLE VII. Spectral reflectance (percent), Style F.

| Marine Pattern (MARPAT) Desert Camouflage Pattern | | | | | | |
|---|---|---|---|---|---|---|
| Wavelength, (nanometers) | Light Tan 479 | | Highland 480 & Light Coyote 481 | | Urban Tan 478 | |
| | Min | Max | Min | Max | Min | Max |
| 700 | 38 | 53 | 19 | 36 | 25 | 48 |
| 720 | 38 | 58 | 20 | 36 | 25 | 52 |
| 740 | 39 | 62 | 20 | 36 | 25 | 54 |
| 760 | 40 | 66 | 21 | 36 | 26 | 56 |
| 780 | 41 | 72 | 21 | 38 | 27 | 57 |



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| Marine Pattern (MARPAT) Desert Camouflage Pattern | | | | | | |
|---|---|---|---|---|---|---|
| Wavelength, (nanometers) | Light Tan 479 | | Highland 480 & Light Coyote 481 | | Urban Tan 478 | |
| 800 | 43 | 76 | 22 | 43 | 28 | 58 |
| 820 | 45 | 76 | 23 | 45 | 30 | 58 |
| 840 | 48 | 78 | 24 | 46 | 33 | 58 |
| 860 | 50 | 78 | 25 | 46 | 36 | 59 |

TABLE VIII. Spectral reflectance (percent), Style G.

| Operation Enduring Freedom Camouflage Pattern (OEF-CP) | | | | | | |
|---|---|---|---|---|---|---|
| Wavelength, (nanometers) | Cream 524 & Tan 525 | | Pale Green 526, Olive 527 & Brown 529 | | Dark Green 528 & Dark Brown 530 | |
| | Min | Max | Min | Max | Min | Max |
| 600 | 22 | 44 | 10 | 30 | 3 | 12 |
| 620 | 24 | 45 | 11 | 30 | 3 | 12 |
| 640 | 24 | 45 | 11 | 32 | 4 | 12 |
| 660 | 25 | 45 | 12 | 32 | 4 | 13 |
| 680 | 28 | 45 | 14 | 35 | 4 | 18 |
| 700 | 28 | 48 | 19 | 40 | 6 | 25 |
| 720 | 30 | 52 | 22 | 43 | 6 | 27 |
| 740 | 32 | 55 | 25 | 46 | 10 | 29 |
| 760 | 36 | 56 | 27 | 48 | 14 | 33 |
| 780 | 38 | 57 | 28 | 50 | 18 | 36 |
| 800 | 40 | 57 | 29 | 50 | 20 | 37 |
| 820 | 44 | 58 | 30 | 51 | 20 | 38 |
| 840 | 46 | 59 | 32 | 51 | 21 | 39 |
| 860 | 48 | 60 | 33 | 52 | 21 | 40 |

TABLE IX. Spectral reflectance (percent), Style H.

| Operational Camouflage Pattern (OCP) | | | | | | |
|---|---|---|---|---|---|---|
| Wavelength, (nanometers) (nm) | Tan 525 & Dark Cream 559 | | Olive 527, Light Sage 560 & Brown 529 | | Dark Green 528 & Bark Brown 561 | |
| | Min | Max | Min | Max | Min | Max |
| 600 | 22 | 44 | 12 | 30 | 3 | 12 |
| 620 | 24 | 45 | 12 | 30 | 3 | 12 |
| 640 | 24 | 45 | 12 | 32 | 4 | 12 |



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| | | | | | | |
|---|---|---|---|---|---|---|
| 660 | 25 | 45 | 12 | 32 | 4 | 13 |
| 680 | 28 | 45 | 14 | 34 | 4 | 18 |
| 700 | 28 | 48 | 14 | 36 | 6 | 25 |
| 720 | 30 | 52 | 16 | 39 | 6 | 27 |
| 740 | 32 | 55 | 18 | 41 | 10 | 29 |
| 760 | 36 | 56 | 20 | 43 | 14 | 33 |
| 780 | 38 | 57 | 22 | 45 | 18 | 36 |
| 800 | 40 | 57 | 22 | 45 | 20 | 37 |
| 820 | 44 | 58 | 24 | 46 | 20 | 38 |
| 840 | 46 | 59 | 26 | 47 | 21 | 39 |
| 860 | 48 | 60 | 28 | 48 | 21 | 40 |

3.8 Physical requirements. The finished cloth shall conform to the requirements specified in Table X when tested as specified in 4.4.5.

TABLE X. Physical requirements.

| Characteristic | Weight (oz/sq yd) | | Yarns per inch (min.) | | Breaking strength lbs (min.) | | Air permeability (cu.ft/min./sq.ft) |
|---|---|---|---|---|---|---|---|
| | Min. | Max. | Warp | Filling | Warp | Filling | Max. |
| Type I Classes 1 and 2 | 8.5 | 9.5 | 35 | 28 | 500 | 300 | 10 1/ |
| Type I Class 3 | 11.0 | 12.0 | 35 | 28 | 500 | 300 | N/A |
| Type I Class 4 | 11.0 | 12.0 | 35 | 28 | 500 | 300 | N/A |
| Type II Classes 1 and 2 | 6.5 | 7.5 | 41 | 36 | 450 | 280 | N/A |



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TABLE X. Physical requirements. - Continued

| Characteristic | Weight (oz/sq yd) | | Yarns per inch (min.) | | Breaking strength lbs (min.) | | Air permeability (cu.ft/min./sq.ft) |
|---|---|---|---|---|---|---|---|
| | Min. | Max. | Warp | Filling | Warp | Filling | Max. |
| Type II Class 3 | 9.0 | 10 | 41 | 36 | 450 | 280 | N/A |
| Type III Classes 1 and 2 | 6.0 | 7.0 | 48 | 35 | 275 | 200 | N/A |
| Type III Class 3 | 7.0 | 8.0 | 48 | 35 | 300 | 225 | N/A |
| Type III Class 4 | 8.0 | 9.5 | 48 | 35 | 360 | 270 | N/A |
| Type IV Classes 1 and 2 | N/A | 5.5 | 58 | 38 | 200 | 155 | N/A |
| Type IV Class 3 | N/A | 6.0 | 58 | 38 | 200 | 155 | N/A |

1/ Requirement applicable to Type I, Class 1 (Solid Shades) only.

3.8.1 Tear strength. The Type IV, Class 2 cloth shall have a tear strength of 8.0 pounds minimum in both the warp and fill. Testing shall be as specified in 4.4.5.

3.8.1.1 Puncture propagation tear strength. The Type IV, Class 3 cloth shall have a tear strength of 7.0 kilograms of force (kgf) minimum in the warp and 6.0 kilograms of force (kfg) minimum in the fill. Testing shall be as specified in 4.4.5.

3.8.2 Abrasion resistance. Unless otherwise specified in the end item use, contract or purchase order (see 6.2) the Type III, Class 4 cloth shall show abrasion resistance to 800 cycles minimum and Type IV, Class 3 shall show abrasion resistance to 700 cycles minimum. Testing shall be as specified in 4.4.5.

3.8.3 Weave. The weave for all Types and Classes shall be plain with one (1) up and one (1) down. The use of a flyshuttle or shuttleless loom is permitted. Testing shall be as specified in 4.4.5.

3.9 Finish. The cloth shall be heat set. Classes 2, 3 and 4 cloths (all Types) shall be given a water repellent treatment (see 3.9.1); Class 3 cloths (all Types) shall be back coated (see 3.9.2) and the Class 4 cloths (Types I and III) shall be flame retardant treated (see 3.9.9).

3.9.1 Water repellency. The Classes 2, 3, and 4 cloths (all Types) shall be given a water



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repellent treatment that shall be capable of meeting all repellency characteristics referenced in this specification.

3.9.1.1 Spray rating. Three (3) individual determinations for Class 2 cloth, shall be equal to or better than 90, 90, 80 initially. Three (3) individual determinations for the Class 3 and Class 4 cloths, shall be equal to or better than 100, 100, 90 initially and 90, 90, 80 after one (1) laundering. Testing shall be as specified in 4.4.5.

3.9.1.2 Hydrostatic resistance. The Class 3 (Types I, II and III), back coated cloths and the Class 4 cloth (Types I and III) shall show no leakage below a hydrostatic height of 35 centimeters. Testing shall be as specified in 4.4.5.

3.9.1.3 Dynamic water absorption (Class 2, 3 and 4). The Class 2 cloth (all Types) shall not have dynamic water absorption greater than 25 percent initially. The cloths for Class 3 (all Types), and Class 4 (Types I and III) shall not show more than 20 percent dynamic water absorption initially and after one (1) laundering. Testing shall be as specified in 4.4.5.

3.9.2 Back coating. The Class 3 cloths (all Types) shall be treated on the back side only with a suitable clear polyurethane coating compound and shall show water repellency characteristics on the face side. If plasticizers are used in the coating, only phosphate or phthalate ester type plasticizers shall be used.

3.9.3 Blocking. The blocking properties at 180°F of the finished back treated side of the Class 3 cloths (all Types) and Class 4 cloths (Types I and III), shall not be greater than a No. 3 rating. Testing shall be as specified in 4.4.5.

3.9.4 Resistance to organic liquid. The finished Classes 2, 3 and 4 cloths (all Types) shall show no wetting by N-Tetradecane minimum initially and after one (1) laundering. Testing shall be as specified in 4.4.5.

3.9.5 Resistance to Diethyltoluamide (DEET). The finished Class 3 cloths (all Types) (and Class 4 cloths (Types I and III) shall show no lifting, tackiness, solution, pickoff or adherence to itself greater than a scale rating of “2” (trace blocking). Testing shall be as specified in 4.4.5.

3.9.6 Resistance to low temperature. The finished Class 3 cloths (all Types), and Class 4 cloths (Types I and III), shall not show any cracking, flaking or separation of the coating from the base cloth. Testing shall be as specified in 4.4.5.

3.9.7 Resistance to high humidity. For the finished Class 3 cloths (all Types), and Class 4 cloths (Types I and III), the coating shall not show stiffness, brittleness, softness or tackiness, and shall show no evidence of cracking or crazing. Testing shall be as specified in 4.4.5.

3.9.8 Stiffness. The stiffness of the Types I, II and III Class 3 finished back treated cloths and Types I and III Class 4 cloths shall not be more than 0.034 pounds per square inch (psi) in the warp or filling directions. The Type IV Class 3 cloth when used for Army uniform components shall



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show stiffness of 0.002 psi (maximum) at 32°F and 70°F when tested as specified Table XI. The Type IV Class 3 cloth when used for Marine Corps uniform components, shall show stiffness of 11 centimeters (maximum) in both the warp and filling when tested as specified in Table XI.

3.9.9 Flame resistance. The Class 4 cloths (Types I and III) shall be flame retardant treated and shall have an average after-flame time of not more than 3.0 seconds in both the warp and fill directions; an average after-glow time of not more than 2.0 seconds in both the warp and fill directions and an average char length of not more than 4-1/2-inches initially and after five (5) launderings in both the warp and fill directions. The average melt/drip after removal of source flame shall be less than one (1) droplet in both the warp and fill directions. Testing shall be as specified in 4.4.5.

3.10 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 or more than 8.5 when tested as specified in 4.4.5.

3.11 Dimensional stability. The finished cloth shall have an average dimensional change of no more than 3.0 percent in the warp and no more than 2.0 percent in the filling directions, with no single value over 3.5 and 2.5 percent, respectively after one (1) laundering cycle. Testing shall be as specified in 4.4.5.

3.12 Width. For Government procurements only, unless otherwise specified, the width of the finished cloth shall be as specified in the contract or purchase order (see 6.2) and shall be the minimum acceptable width inclusive of selvages. When the cloths are woven on shuttleless looms, the width measurement shall be made between the last yarns on each side, with the pro- truding fringe(s) excluded.

3.13 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished full width in continuous lengths, each not less than 40-yards.

3.14 Face identification. The face side of Style A, solid shade dyed cloth shall be identified by stamping that side with the word “FACE” at each end of the roll.

3.15 Roll identification. Each roll of finished cloth shall be labeled or ticketed for fiber con- tent in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

3.16 Toxicity The finished fabric shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.4.5.
Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.17 Workmanship. The finished cloth shall conform to the quality of product established by this document. The demerit points per 100 square yards when calculated as specified in Section 4 shall not exceed the established maximum point value.



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## 4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance, color and finish defects in 4.4 and tested for the characteristics specified in 4.4.5 through 4.5.7 as applicable.

4.2.1 First article samples. Unless otherwise specified in the contract or purchase order, the sample unit shall be one (1) continuous 5 yard length of full width, finished cloth.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the testing in 4.4.5 through 4.5.7 as applicable.

4.3.1 Conformance inspection samples. Sampling for finished cloth testing shall be performed in accordance with the following lot sample size for yards of cloth:

| Lot size (yards) | Sample size (sample units) |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5. |

4.3.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with all the requirements of referenced documents, unless otherwise, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2).

4.4 Inspection methods.

4.4.1 Yard by yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in Section III of FED-STD-4, that are clearly noticeable at normal inspection distance (3-feet) shall be scored and assigned demerit points as listed in 4.4.1.1 except that only those slubs and knots which exceed the limits shown on the Fabric Defect Scale (see 2.3), "D" for slubs and "C" for knots, shall be scored and coarse yarn shall only be scored as a defect when the coarse yarn is twice the diameter of the normal yarn used in the fabric. No linear yard (increments of 1-yard on the measuring device of the inspection machine) from any one (1) roll shall be penalized more than four (4) points. The sample size shall be 20 rolls selected from 20 containers. No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present



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in the sample. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceed 35.0 points. The lot shall be unacceptable if the points per 100 square yards of two (2) or more individual rolls exceed 53.0 points. If one (1) roll in the sample exceeds 53.0 points per 100 square yards, a second sample of 20 rolls shall be examined for individual roll quality only. The lot shall be unacceptable if one (1) or more rolls in the second sample exceeds 53.0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be calculated as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.4.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to and including 3-inches in any dimension - one (1) point
For defects exceeding 3-inches, but not exceeding 6-inches in any dimension - two (2) points
For defects exceeding 6-inches, but not exceeding 9-inches in any dimension - three (3) points For defects exceeding 9-inches in any dimension - four (4) points

4.4.1.2 Four (4) demerit point defects. The following shall be scored four (4) points for each yard in which they occur:

Hole, cut or tear
Objectionable odor
Baggy, ridged or wavy cloth
Width less than specified
Edge ravels when pulled outward
Slack or tight selvage 1/
Overall uncleanness
Pattern design not equal to standard sample (Styles B, C, D, E, F, G and H)
Incorrect color in any part of the pattern (Styles B, C, D, E, F, G and H)
Pattern repeat not equal to the standard sample (Styles B, C, D, E, F, G and H)
Pattern repeat in both the warp and filling directions less or more than the specified dimensions (Styles B, C, D, E, F, G and H)
Skitteriness (mottled, uneven color) of pattern exceeds that shown by standard sample (Styles B, C, D, E, F, G and H)
Excessive feathering or spew (fuzziness at color boundaries) of pattern as compared to the standard sample (Styles B, C, D, E, F, G and H)
Excessive grinning (off register, gap where ground shade shows through) of pattern as compared to the standard sample (Styles B, C, D, E, F, G and H)
Excessive haloing or trapping (overlapping of colors) of pattern as compared to the standard sample (Styles B, C, D, E, F, G and H)



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1/ To determine the presence of unacceptable selvage conditions, the following procedure shall be followed: During the visual examination, the perch shall be stopped a minimum of three (3) times for each roll in the sample, tension removed and the finished cloth examined for selvage conditions. A waviness in the selvage or significant ripples diagonally across the width of the fabric is an indication of slack or tight selvage.

4.4.1.3 Examination (treated fabric) Class 3. In addition to the defects listed in 4.4.1.1 the required yardage of each roll of the finished treated cloth shall be inspected on the treated side for the visual defects listed below and as defined in MIL-STD-1487. The defects found shall be counted regardless of their proximity to each other, except where two (2) or more defects represent a single local condition on the cloth, in which case only one (1) defect shall be counted. A continuous defect shall be counted as one (1) defect for each warpwise yard in which it occurs. The following shall be scored four (4) points for each yard in which they occur.

| Examine | Defect |
|---|---|
| Coating | Any uncoated area, Any thinly coated area Any blister, tunnel, or delamination of coating, Any lump or heavily coated area |
| | Crease or wrinkle that cannot be corrected by manual pressure or resulting in doubling or adhesion of surfaces |
| | Any spot, stain, or streak more than 1-inch in its longest dimension 1/ |
| | Any embedded foreign matter |
| | Any scorch or burn |
| | Any strike through of the coating to the uncoated side of the cloth Tackiness 1/ Clearly visible at normal inspection distance (approximately 3 feet). |

4.4.2 Length examination. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2-yards less than the length marked on the ticket shall be considered a defect. The lot shall be unacceptable if two (2) or more rolls in the sample are defective in respect to length. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.4.3 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and appearance. Any roll in the sample, off shade or shaded side to side, side to center, or end to end, or any roll that does not have the same appearance as the standard sample, shall be cause for rejection of the entire lot.

4.4.4 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for the defects listed below. The lot shall be unacceptable if two (2) or more of the following defects are present in any sample:

Not labeled or ticketed in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act



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Face marking missing from either or both ends (Style A, solid shade)
Face marking on wrong side (Style A, solid shade)

4.4.5 End item testing. The cloth shall be tested for characteristics listed in Table XII. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 5-yards, full width of finished cloth. The sample size shall be as follows and the lot shall be unacceptable if one (1) or more sample units fail to meet any requirement.

| Lot size (yards) | Sample size (sample units) |
|---|---|
| 800 or less | 2 |
| 801 up to 22,000 | 3 |
| 22,001 and over | 5 |



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TABLE XI. End item tests.

| Characteristic | Requirement Paragraph | Test Method |
|---|---|---|
| Yarn: | | |
| Fiber identification | 3.4.1 | AATCC-20 or ASTM D276 |
| Yarn denier | 3.4.1.1 | ASTM D1907/D1907M |
| Visual shade matching | 3.5.9 | 1/ |
| Colorfastness to: | | |
| Laundering (after three (3) cycles) | 3.5.10 & Table I | AATCC-61, Test 1A 2/ |
| Light (after 40 AFU or 170 kJ/ ($m^2$nm)@ 420 nm) | 3.5.10 & Table I | AATCC-16.2 or 16.3 3/ |
| Perspiration (acid & alkaline) | 3.5.10 & Table I | AATCC-15 2/ |
| Crocking (wet and dry) | 3.5.10 & Table I | AATCC-8 4/ |
| Accelerated Weathering (after 80 AFU or 340 kJ/($m^2$nm)@ 420 nm (Style A only) | 3.5.10 & Table I | AATCC-169 Option 3 3/ |
| Frosting (carbon black only) (300 cycles) | 3.5.10 & Table I | AATCC-119 3/ |
| Pattern Execution | 3.6 | Visual |
| Spectral reflectance | 3.7 and Tables II- IX | 4.5.1 |
| Weight | 3.8 & Table X | ASTM D3776/D3776M, Option C |
| Yarns per inch | 3.8 & Table X | ASTM D3775 |
| Breaking strength | 3.8 & Table X | ASTM D5034 |
| Air permeability (Type I Class 1 only) | 3.8 & Table X | ASTM D737 |
| Tear strength: (Type IV Class 2) | 3.8.1 | ASTM D1424 |
| Puncture propagation tear strength (Type IV Class 3) | 3.8.1.1 | ASTM D2582 |
| Abrasion Resistance | 3.8.2 | ASTM D3884 5/ |
| Weave | 3.8.3 | Visual |
| Spray rating | | |
| Initial: All Types (Classes 2 & 3) and Types I and III (Class 4) | 3.9.1.1 | AATCC-22 |
| After one (1) laundering All Types (Classes 2 & 3) and Types I and III (Class 4) | 3.9.1.1 | AATCC-96 Test VIc,A,0 and AATCC-22 |



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TABLE XI. End item tests. - Continued

| Characteristic | Requirement Paragraph | Test Method |
|---|---|---|
| Hydrostatic resistance<br>Types I, II & III (Class 3)<br>and Types I and III (Class 4) | 3.9.1.2 | AATCC-127 or ASTM D751, 6/, 7/ |
| Dynamic absorption:<br>Initial: All Types (Classes 2 & 3)<br>and Types I and III (Class 4)<br>After one (1) laundering<br>All Types (Classes 2 & 3)<br>and Types I & III (Class 4) | 3.9.1.3<br><br>3.9.1.3 | AATCC-70<br><br>AATCC-96, Test VIc,A,0 and AATCC-70 |
| Blocking (All Types, Classes 3 & 4) | 3.9.3 | ASTM D751, and 4.5.2 |
| Resistance to organic liquid:<br>Initial: All Types (Classes 2 & 3)<br>and Types I & III (Class 4)<br>After one (1) laundering<br>All Types (Classes 2 & 3)<br>and Types I & III (Class 4) | 3.9.4<br><br>3.9.4 | AATCC-118 8/<br><br>AATCC-96, Test VIc,A,0 and AATCC-118 |
| Resistance to Diethyltoluamide (DEET) | 3.9.5 | 4.5.3 |
| Resistance to low temperature<br>All Types (Class 3)<br>and Types I & III (Class 4) | 3.9.6 | 4.5.4 |
| Resistance to high humidity | 3.9.7 | 4.5.5 |
| Stiffness:<br>Types I, II & III (Class 3) @ 70°F<br>and Types I & III (Class 4) @ 70°F<br>Type IV (Class 3) Army @ 32 °F& 70°F<br>Type IV (Class 3) Marine Corps | <br><br>3.9.8<br>3.9.8<br>3.9.8 | <br><br>ASTM D747<br>ASTM D747<br>4.5.6 |
| Flame retardant (Class 4 Only)<br>Initial<br>After five (5) launderings | 3.9.9<br>3.9.9 | ASTM D6413/D6413M 9/<br>AATCC-135, 3,III,Aiii &<br>ASTM D6413/D6413M 9/ |
| pH | 3.10 | AATCC-81 |
| Dimensional stability(after 1 cycle) | 3.11 | AATCC-96, Test VIc,A,0 |
| Toxicity | 3.16 | 4.5.7 |

1/ The color and appearance of the cloth shall match the standard sample when viewed using the AATCC Evaluation Procedure 9, Option A, with sources simulating artificial daylight D75 illuminant with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles, and shall be a



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good match to the standard sample under incandescent A illuminant with a color temperature of 2856K.
2/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change and AATCC Evaluation Procedure 2, Gray Scale for Staining.
3/ Rated using the AATCC Evaluation Procedure 1, Gray Scale for Color Change
4/ Rated using the AATCC Evaluation Procedure 8, AATCC 9-Step Chromatic Transference Scale
5/ H-18 abrasive wheel with 1000 gram load shall be used. A hole shall be defined as the wear through of one (1) warp end and one (1) filling yarn at the same location.
6/ Leakage is defined as the appearance of water at three or more different places within the 4-1/2-inch diameter test area at a hydrostatic height of 35.0 centimeters. The uncoated side of the coated cloth shall contact the water.
7/ In cases of dispute, the ASTM method prevails
8/ Test for N-Tetradecane minimum only.
9/ Specimen which burn along the edge of the specimen holder shall be considered invalid and retested.

4.5 Methods of inspection.

4.5.1 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometers (nm) for Woodland Camouflage (Styles B and E), Universal Camouflage (Style D), Operational Enduring Freedom Camouflage (OEF-CP - Style G) and Operational Camouflage (OCP – Style H) and most solids, 700 to 860 nanometers (nm) for Desert Camouflage (Styles C and F) at 20 nm intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white or vitrolite tiles. The spectral band width shall be less than 20 nm at 860 nm. Reflectance measurements shall be made by either the mono-chromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either CIE Source A or CIE Source D65. Measurements shall be taken on a minimum of two (2) different areas and the data averaged. The measured areas should be at least 6-inches away from the selvage. The specimen shall be measured as a single layer backed with layers of the same fabric and shade as follows: Styles B and E (Woodland print cloths) three (3) backing layers of the same shade for Light Green 354, Dark Green 355 and Brown 356 colors and two (2) backing layers of the same shade for Black 357. Styles C and F (Desert print cloths) four (4) layers of the same shade cut from the standard. Style D (Universal Camouflage print), Style G (OEF-CP) and Style H (OCP) with four (4) layers of the same shade. The specimen shall be viewed at an angle no greater than 10° from normal, with the specular component included. Measurements shall be taken on a minimum of two (2) different areas. Specimens shall be oriented in different directions during testing. When possible, the specimens tested shall not contain the same yarns when presented to the sample port. Camouflage materials should be measured with the appropriate aperture size to ensure that only one (1) color is measured at a time. (NOTE: Always use the largest aperture possible). The diameter for standard aperture size used in the color measurement device shall be 1.0 to 1.25-inches for most solid colors (Style A), Woodland (Style B) and Desert



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Camouflage (Style C) and 0.3725-inches or larger for the Universal Camouflage (Style D), Woodland MARPAT (Style E), Desert Camouflage MARPAT (Style F), OEF-CP (Style G), and OCP (Style H). Photometric accuracy of the spectrophotometer shall be within one (1) percent and wavelength accuracy within two (2) nm. Any color having spectral reflectance values falling outside the limits at four (4) or more of the wavelengths specified shall be considered a test failure.

4.5.2 Blocking. The test shall be performed in accordance with ASTM D751, Blocking Resistance at Elevated Temperatures, except that the test shall be performed at a temperature of 180 (± 2) ºF for 30 minutes. Evaluate the resistance of the specimen to blocking by the scale given below:

1 – No Blocking. Cloth surfaces are free and separate without any evidence of cohesion or adhesion.
2 – Trace blocking. Cloth surfaces show slight cohesion or adhesion.
3 – Slight blocking. Cloth surfaces must be lightly peeled to separate.
4 – Blocking. Cloth surfaces separate with difficulty or coating is removed during separation.

4.5.3 Resistance to Diethyltoluamide (DEET). The DEET solution contains 75 percent diethyltoluamide and 25 percent ethanol (see 4.5.3.1). Three (3) drops of the DEET solution shall be placed in the center of a 4 by 8-inch specimen of the finished cloth with the coated side up. The specimen shall be folded to form a 4 by 4-inch square with the coated side folded onto itself. The folded specimen shall then be placed between two (2) 6 by 6-inch glass plates and a 4-pound weight placed on the assembly and left at standard conditions for 16 hours. The specimen shall then be removed from between the glass plates, and immediately rated using the blocking scale ratings as shown in 4.5.2.

4.5.3.1 DEET reagent. The DEET reagent is an insect repellent reagent solution of 75 percent by weight (min) of diethyltoluamide and the remainder denatured alcohol. The diethyltoluamide component of the solution shall be a technical grade and containing N, N-diethylmetatoluamide of not less than 95 percent purity and the remainder shall consist of entirely or mixture of ortho or para isomers of N, N- diethyltoluamide. The denatured alcohol component of the solution shall be ethanol, U.S.P. 94.9 percent by volume and denatured in accordance with, The Code of Federal Regulations 27 CFR 21, Formula 40 (see 2.2.2). The diethyltoluamide shall be registered with the U. S. Environmental Protection Agency in accordance with the Federal Insecticide, Fungicide and Rodenticide Act (FIFRA) (see 2.2.2).

4.5.4 Resistance to low temperature. The test shall be performed in accordance with ASTM D751, Low Temperature Crack Test, with the exposure time 4-hours (min.) at a test temperature of -40°F (± 5)°F; the test for hydrostatic resistance shall not be performed. The specimen shall be removed from the chamber, allowed to come to room temperature and visually examined for any signs of cracking, flaking or separation of the coating from the base cloth. Unless otherwise specified, at least three (3) specimens from the sample shall be tested. Results of tests shall be expressed as “pass” or “fail” as exhibiting visible coating nonconformities.



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4.5.5 Resistance to high humidity. Three 4 by 4-inch specimens shall be laid flat, coated side up, on a supporting plate and the assembly placed in a desiccator containing water in the lower portion. The water level shall be approximately 1-inch below the specimens. The lid of the desiccator shall be put in place and the desiccator placed in a circulating air oven having a temperature of 125°F (± 2)°F for a period of seven (7) days. At the end of the aging period, each specimen shall be removed from the desiccator, visually examined for colorfastness and then visually examined for any evidence of stiffness, brittleness, softness, tackiness, cracking or crazing.

4.5.6 Stiffness test.

Apparatus. The test apparatus shall consist of a framework with a rotating, two (2) roller spindle assembly and a calibrated linear measuring tool. Of the assembly, the two (2) rolls, each 1-inch in diameter and approximately 4.25-inches in length, are positioned parallel to one another and held in contact by spring tension. The line of contact of the two (2) rolls shall coincide with the axis of rotation of the spindle assembly attached to the fixed framework. A pointer shall be attached to the spindle assembly to indicate the relative angular position of the assembly to a moveable circular scale calibrated in degrees. The rolls of the spindle assembly shall be capable of being rotated on their axis using a slow gear adjustment to adjust the length of the test specimen. The assembly shall rotate, in both clockwise and counterclockwise directions, at a uniform rate of one (1) revolution per 60 (± 5) seconds. The linear measuring tool, graduated to 0.1 mm, shall be used to measure the length of a test specimen extending perpendicular from the line of contact or nip of the two (2) rolls.

Test specimens. The specimen shall be a rectangle of cloth 1-1/4-inches wide by 6- to 12-inches long with the longer dimension parallel to the direction being tested; unless otherwise specified, the warp or machine direction of the sample shall be tested. The specimens shall be cut with clean, straight and parallel edges from locations diagonally across the width of the sample and they shall not contain any evidence of creasing or folding.

Number of determinations. Unless otherwise specified, five (5) specimens from each of the designated directions shall be tested from each sample unit.

Procedure. Conditioned test specimens in accordance with ASTM D1776/D1776M and test with specimens and apparatus in that environment, unless otherwise specified. Level the apparatus, before use, so that the spindle assembly is horizontal. Secure an end of the test specimen in and perpendicular to the nip of the rolls with enough length projecting on the left so that on rotating the spindle assembly clockwise, the projecting end of the test specimen falls through the vertical to the right. On rotating the spindle assembly counterclockwise from the stop or end point, the test specimen should not fall back to the left until it is turned through an angle of 90 degrees. Shorten the projecting length until a rotation of 90 (± 2) degrees causes the end of the test specimen to fall from one side to the other. This defines the critical length which is measured from the line gripping the specimen (or nip of the rollers) to its free end. Measure the critical length of the test specimen in millimeters and to the nearest millimeter using the linear measuring tool. Report all individual



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readings of critical length and the average of results for the specified direction of test of each sample unit.

4.5.7 Toxicity test. Unless otherwise specified (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of the studies indicate the finished cloth is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If the toxicity requirement (see 3.16) can be demonstrated with historical use data, toxicity testing may not be required (see 6.2).

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory or contractually binding.)

6.1 Intended use. The cloth is intended for use in the manufacture of load bearing vests, field packs, body armor protective vests, duffel bags, reinforcement elbow and knee patches and other field items.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this Specification.
b. Type, Class and Style of cloth required (see 1.2).
c. The specific issue of individual documents referenced (see 2.2 and 2.3).
d. When first article is required (see 3.1, 4.2 and 6.3).
e. Color required if Style A is specified (see 3.5.1).
f. Pattern drawing or standard sample, if printed camouflage required (see 3.6).
g. When abrasion requirement is not applicable (see 3.8.2)
h. Width of cloth required, (see 3.12).
i. Length required if other than specified (see 3.13).
j. When toxicity testing if required (see 3.16 and 4.5.7).
k. Conformance inspection conditions (see 4.3)
l. Inspection conditions (see 4.3.1).



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m. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required (see 3.1), it will be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a pre-production sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Toxicity. Further information on chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens may be found at http://www.epa.gov/ebtpages/pollutants.html.

6.6 Supersession data . This document supersedes MIL-C-43734D. The supersession data is as follows:

| MIL-C-43734D | MIL-DTL-32439 |
|---|---|
| Class 1 | Type I, Class 1, Style A |
| Class 2 | Type I, Class 2, Style B |
| Class 3 | Type I, Class 3, Style A |
| Class 4 | Type III, Class 3, Style B |
| Class 5 | Type III, Class 2, Style B |

6.7 Subject term (key word) listing).

Vest, Survival, AIRSAVE
Bag, Duffel
Camouflage, Desert,
Camouflage, MARPAT
Camouflage, Woodland
Camouflage, Universal

Camouflage, OEF-CP
Camouflage, OCP
Equipage item
Flame retardant
Water repellency

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of changes.



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Custodians:
Army – GL
Navy – NU
Air Force – 11

Preparing Activity:
DLA-CT

Review Activities:
Army – MD, AV
Navy – MC, AS

(Project 8305-2017-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at https://assist.dla.mil/.



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